Prompts to Practice Using the Coast Formula

Prompt: Act as a Cisco CCNA Instructor. Create ten well-constructed prompts relating to IT (6 beginner-level, 3 intermediate-level, and 2 advanced-level) that students can use to break it down into the five elements of the C.O.A.S.T prompting formula. Think deeply about this.
Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output.
Here is an example:
'''
This example demonstrates how each component of the C.O.A.S.T. formula contributes to creating a clear, actionable prompt to guide the AI in generating relevant and helpful responses.
Prompt: "You are a marketing strategist working for a tech startup launching a new smartphone app. Research other marketing campaigns and analyze target market demographics, identify key marketing channels, create a messaging framework, design a promotional campaign, and set up tracking metrics.
Find similar apps in the market that have succeeded through social media marketing and influencer partnerships. Create a detailed marketing plan geared toward online and printed consumers. Include budget allocation, timeline, and success metrics for the app launch."
Context: You are a marketing strategist working for a tech startup launching a new smartphone app.
Objective: Research other marketing campaigns:

- Analyze target market demographics
- Identify key marketing channels
- Create a messaging framework
- Design a promotional campaign
- Set up tracking metrics

Audience: Consumers of online and print media
Scenario: Find similar market apps that have succeeded through social media marketing and influencer partnerships.
Task: Create a detailed marketing plan including budget allocation, timeline, and success metrics for the app launch.
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Internal Rubric for a "World-Class" Answer

Before constructing the prompts, I developed the following internal rubric to define a 'world-class' response:

1. **Alignment with COAST**: Each prompt must be clearly decomposable into the five COAST elements (Context, Objective, Actions, Scenario, Task), with each element explicit and meaningful.
2. **Relevance to CCNA/IT**: Prompts must be directly relevant to core CCNA and broader IT concepts, reflecting current curriculum and industry needs.
3. **Clarity and Specificity**: Prompts must be unambiguous, detailed, and tailored to the target audience (students preparing for CCNA).
4. **Cognitive Demand and Scaffolding**: Prompts should span a range of difficulty levels, from foundational to advanced, and encourage both theoretical understanding and practical application.
5. **Completeness**: Each prompt should cover all necessary aspects of the topic, ensuring comprehensive engagement with the material.

6. **Creativity and Problem-Solving**: Prompts should foster critical thinking, synthesis, and real-world problem-solving.
7. **Actionability**: Each prompt should result in a tangible, assessable output or deliverable.
8. **Diversity of Topics**: The set of prompts should collectively cover a broad spectrum of CCNA and IT domains (e.g., networking, security, automation, wireless, troubleshooting, etc.).
9. **Pedagogical Soundness**: Prompts should reflect best practices in instructional design, including active learning and hands-on engagement.
10. **Professional Tone and Style**: Prompts should maintain a professional, supportive, and instructional tone appropriate for technical education.

1. Network Topology Design for a Small Business

Prompt: You are a network engineer tasked with designing a reliable and scalable network for a small business with 50 employees. Research best practices for small office network design, select appropriate network devices, create a logical topology diagram, justify your design choices, and outline a plan for future expansion.

- **Context:** You are a network engineer for a small business with 50 employees.
- **Objective:** Design a reliable and scalable network.
- **Actions:** Research best practices, select devices, create a topology diagram, justify choices, and plan for expansion.
- **Scenario:** The business anticipates growth and needs a network that can scale.
- **Task:** Deliver a logical network topology diagram with a written justification and an expansion plan.

2. Securing Network Devices Against Common Threats

Prompt: As a junior network administrator, you are responsible for securing the company's routers and switches. Identify common security threats, research device hardening techniques, implement at least three security configurations on Cisco devices, test their effectiveness, and document your process.

- **Context:** You are a junior network administrator securing routers and switches.
- **Objective:** Protect network devices from common security threats.
- **Actions:** Identify threats, research hardening techniques, implement configurations, test, and document.
- **Scenario:** The company has recently experienced attempted unauthorized access.
- **Task:** Submit a configuration report and a summary of security measures taken.

3. Troubleshooting Inter-VLAN Routing Issues

Prompt: You are called to troubleshoot a network where devices on different VLANs cannot communicate. Analyze the current configuration, identify possible misconfigurations, propose corrective actions, implement changes in a lab environment, and verify successful inter-VLAN communication.

- **Context:** Devices on different VLANs cannot communicate.
- **Objective:** Restore inter-VLAN connectivity.
- **Actions:** Analyze configuration, identify issues, propose fixes, implement, verify.
- **Scenario:** The network uses Layer 3 switches for inter-VLAN routing.

- **Task:** Provide a troubleshooting report detailing the issue, solution, and verification steps.

4. Implementing DHCP and NAT in a Branch Office

Prompt: You are deploying a new branch office network that requires dynamic IP addressing and internet access. Research DHCP and NAT concepts, configure a Cisco router to provide DHCP services and NAT for internet connectivity, test the setup, and explain how each configuration supports branch operations.

- **Context:** New branch office needs dynamic IPs and internet access.
- **Objective:** Implement DHCP and NAT for the branch.
- **Actions:** Research concepts, configure the router, test, and explain configurations.
- **Scenario:** The branch has 30 users and a single internet connection.
- **Task:** Submit configuration files and a written explanation of your setup.

5. Wireless Network Planning and Security

Prompt: You are designing a wireless network for a university campus. Assess coverage requirements, select appropriate Wi-Fi standards, plan access point placement, implement security protocols, and create a deployment guide for IT staff.

- **Context:** University campus wireless network design.
- **Objective:** Provide secure, reliable wireless coverage.
- **Actions:** Assess requirements, select standards, plan placement, implement security, and create a guide.
- **Scenario:** The campus includes classrooms, dorms, and outdoor areas.
- **Task:** Deliver a deployment guide with diagrams and security recommendations.

6. Automating Network Configuration with Python

Prompt: As part of a network automation initiative, you are asked to automate the configuration of multiple Cisco switches. Research Python scripting for network automation, write a script to configure VLANs on several devices, test the script in a simulated environment, and document your code and results.

- **Context:** Automating switch configuration using Python.
- **Objective:** Streamline VLAN configuration across multiple devices.
- **Actions:** Research scripting, write script, test, document.
- **Scenario:** The organization manages dozens of switches with frequent VLAN changes.
- **Task:** Submit your Python script, test results, and documentation.

7. Analyzing Network Traffic with Wireshark

Prompt: You are investigating intermittent connectivity issues in a corporate network. Use Wireshark to capture and analyze network traffic, identify patterns or anomalies, correlate findings with theoretical protocol behavior, and recommend solutions to resolve the issues.

- **Context:** Corporate network experiencing intermittent connectivity problems.
- **Objective:** Diagnose and resolve network issues using traffic analysis.
- **Actions:** Capture traffic, analyze, identify anomalies, correlate with theory, recommend solutions.
- **Scenario:** Issues occur during peak business hours.

- **Task:** Provide an analysis report with findings and actionable recommendations.

8. Designing a Secure Remote Access Solution

Prompt: Your company needs to enable secure remote access for remote employees. Research VPN technologies, compare SSL and IPsec VPNs, design a remote access solution, configure a test environment, and evaluate the security and usability of your implementation.

- **Context:** The Company requires secure remote access for employees.
- **Objective:** Design and implement a secure VPN solution.
- **Actions:** Research, compare technologies, design, configure, evaluate.
- **Scenario:** Employees use a mix of company-issued and personal devices.
- **Task:** Submit a design document, configuration steps, and evaluation summary.

9. Monitoring and Managing Network Performance

Prompt: You are responsible for monitoring network performance in a medium-sized enterprise. Identify key performance metrics, select appropriate monitoring tools (e.g., SNMP, NetFlow), set up monitoring, analyze collected data, and propose optimizations to improve network efficiency.

- **Context:** Monitoring network performance in an enterprise environment.
- **Objective:** Ensure optimal network performance through monitoring and analysis.
- **Actions:** Identify metrics, select tools, set up monitoring, analyze data, and propose optimizations.
- **Scenario:** The network supports critical business applications with strict uptime requirements.
- **Task:** Deliver a monitoring setup report and a list of recommended optimizations.

10. Integrating Cloud Services into Enterprise Networks

Prompt: Your organization is migrating some services to the cloud. Evaluate the impact of cloud integration on network design, identify necessary changes to routing and security, develop a migration plan, implement a pilot integration, and assess the results.

- **Context:** Organization migrating services to the cloud.
- **Objective:** Integrate cloud services while maintaining network security and performance.
- **Actions:** Evaluate impact, identify changes, develop a plan, implement a pilot, and assess results.
- **Scenario:** The migration includes email, file storage, and collaboration tools.
- **Task:** Submit a migration plan, pilot results, and recommendations for full deployment.